

Manish Vemula
Machine Learning Engineer

+1 (734) 262-8577 | Manish.vs145@gmail.com | [LinkedIn](#)

Machine Learning Engineer experienced in building real-time data pipelines, developing ML/NLP models, and integrating Generative AI components using Python, Kafka, PySpark, and Scikit-learn. Skilled in feature engineering, risk and anomaly models, model monitoring, and workflow automation. Background includes conversational AI, lead scoring, sentiment analysis, and CRM data workflows. Certified in AWS Machine Learning and Azure Data Science with strong academic training in Information Systems and Business Administration.

EXPERIENCE

ML Engineer (AI) | **Discover Financial Services, TX – USA** **September 2024 – Present**

- Assisted in setting up real-time data pipelines to collect live customer and transaction events using Kafka, SQL, and Python.
- Helped create and maintain feature pipelines that transform raw streaming data into machine-learning features using PySpark, Pandas, and data engineering basics.
- Supported the development and testing of ML models for risk scoring and anomaly detection using Scikit-learn, XGBoost, model evaluation, and Python.
- Worked with the team to integrate a Generative AI (LLM) component using prompt engineering, embeddings, and NLP techniques.
- Contributed to building a decision engine using Python, REST APIs, and basic rule-based logic to combine model outputs with business rules.
- Helped deliver a real-time customer decision system by collaborating with teams using Git, Jira, Agile practices, and CI/CD basics.
- Assisted in monitoring model performance and checking data quality using ML model monitoring, logging tools, and basic statistical analysis to ensure reliability and compliance.

Associate Machine Learning Engineer | **Nexova – India** **May 2020 – July 2022**

- Helped build an AI system that handles customer calls/chats using speech-to-text, NLP basics, and Python scripting.
- Helped clean and preprocess conversation data using tokenization, text normalization, and basic data cleaning techniques to improve model accuracy.
- Assisted in creating an automated appointment booking flow using API integration and workflow automation.
- Set up data pipelines to store call transcripts in CRM using ETL concepts, JSON handling, and database basics.
- Supported development of a basic lead-scoring model using scikit-learn, feature extraction, and model evaluation.
- Performed simple customer sentiment checks using text classification and natural language processing.
- Added fallback rules using rule-based logic, confidence thresholds, and error handling.
- Helped create CRM dashboards using data visualization, reporting tools, and SQL queries.
- Assisted in generating simple marketing content using text generation tools, prompt engineering, and generative AI basics.
- Helped fine-tune small NLP models on customer conversation data using Hugging Face, data labeling, and model training basics.
- Worked on recommending content types using basic analytics, engagement metrics, and prediction logic.

SKILLS

Programming & Automation: Python, REST API, Automation scripts
Machine Learning & AI: ML algorithms, Feature engineering, Model testing, NLP, Sentiment analysis, Generative AI, LLM prompt engineering, Hugging Face
Big data & Distributed Systems: PySpark, Kafka, Basic distributed computing concepts
Data Engineering & Pipelines: Kafka, ETL, PySpark, Pandas, SQL, JSON, CRM data workflows
MLOps & Deployment: Git, GitHub, CI/CD, Model monitoring, Logging
Data Visualization: Dashboard creation, Reporting tools, Power BI, Tableau, CRM dashboards
Collaboration & Agile: Agile, Scrum, Documentation, Jira
Cloud platforms: AWS (SageMaker, S3, Lambda, EC2), Terraform, Azure ML

EDUCATION

Master of Business Administration (MBA) Central Michigan University, Mount Pleasant, MI	2024 – 2025
Masters of Science in Information Systems Central Michigan University, Mount Pleasant, MI	2022 – 2024
Bachelor of Computer Science Jawaharlal Nehru Technological University, Hyderabad, IND	2017 – 2021

CERTIFICATIONS

Microsoft Certified: Azure Data Scientist Associate
AWS Certified Machine Learning - Specialty